Markowitz Mean-Variance Portfolio Optimization

An Application to Commodity Exchange-Traded Funds

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Abstract

This research examines Markowitz's mean-variance portfolio optimization on ten commodity exchange-traded funds from January 2020 to December 2025. The basic idea is that the risk of the portfolio depends not only on the risk of each asset, but also on how their movements are correlated. Thus the problem reduces to a straightforward application of probability and linear algebra in general. Expected return can be expressed as a vector. The relationships between asset returns can be expressed as a covariance matrix. Portfolio variance can be expressed as a quadratic expression. I derive the equations for the expected portfolio return, variance, and the global minimum-variance portfolio. I then evaluate these metrics using historical ETF data. And finally, I create a long only efficient frontier and a maximum sharpe portfolio. The paper emphasizes the importance of covariance in diversification and makes clear that the portfolio with the lowest variance is not always the portfolio with the best risk-adjusted return.

1 Introduction

1.1 Motivation

As an incoming junior studying Operations Research: Financial Engineering at Columbia University. I've built projects in the past on options pricing and crypto trading. I wanted to use this project to connect the course material to a subject I am passionate about.

1.2 Overview

Investing is always a balancing act between return and risk. Generally, an investor prefers a higher expected return. But higher expected returns may also come with higher volatility (In finance, volatility is another term for risk). The interesting challenge is not simply to find an investment that gives good returns or one that has a low risk. The challenge is determining the optimal mix of assets to get the most advantageous tradeoff between risk and return. Harry Markowitz formalized this idea in mean-variance portfolio theory [1, 2]. Within the Markowitz framework, the portfolio is evaluated as a whole, rather than only an asset's expected return and variance, taking into account the inter dependencies across asset returns which can be denoted through the covariance matrix. If two investments are correlated, then combining them will provide little diversification benefit. If they move in different directions, their profits and losses may cancel out somewhat, thus lowering the variance of the entire portfolio. This topic is relevant to both parts of MATH 2015 as the financial math involved in the paper, are mostly from this course. Asset returns can be modeled as random variables. Expected value describes their average behavior, variance measures dispersion and covariance and correlation describe the relationship between pairs of returns. These ideas were developed directly in the probability strand of the curriculum [5, 6]. Linear algebra naturally represents a portfolio as a vector of weights. The matrix represents the relationships of all asset returns. Expected return and variance can be represented compactly through matrix multiplication and matrix inversion features in the global minimum-variance portfolio equation [7, 8]. For the empirical part of the project, I use ten commodity ETFs:

GLD, IAU, SLV, PPLT, USO, BNO, DBA, CORN, CPER, UNG.

The choice of investments includes gold, silver, platinum, crude oil, agricultural commodities, copper and natural gas. I picked commodities because the assets are related but not identical. For example, GLD and IAU both track gold and should be highly correlated, whereas gold and natural gas are driven by completely different markets. This allows the dataset to be useful for investigating the effect of covariance on an optimized portfolio. The purpose of the study is to develop a classroom-based mean variance framework, to estimate the model with historical data, to construct the efficient frontier, and to analyze the min-variance and the max-risk-adjusted return portfolios.

2 Mathematical Framework

Before delving into the research, it is important we first acknowledge the math used in this project.

2.1 Asset Returns as Random Variables

Let R_i denote the return on asset i. Because the future value of R_i is uncertain, it can be modeled as a random variable. Its expected return is

$$\mu_i = E[R_i].$$

In the empirical part of this project, the true expected return is unknown, so I estimate it using the historical sample mean. This follows the same idea as using observed data to estimate the expected value of a random variable.

Risk is represented by variance. For a return R_i,

$$\mathrm{Var}(R_i) = E[(R_i - \mu_i)^2].$$

The standard deviation is

$$\sigma_i = \sqrt{\mathrm{Var}(R_i)}.$$

In financial applications, this standard deviation of returns is usually called volatility.

When several assets are considered at once, the relationships between them also matter. For two returns R_i and R_j,

$$\mathrm{Cov}(R_i, R_j) = E[R_i R_j] - E[R_i]E[R_j].$$

The course notes define covariance in this way and also emphasize that zero covariance does not necessarily imply independence. The corresponding correlation coefficient is

$$\rho_{ij} = \frac{\mathrm{Cov}(R_i, R_j)}{\sigma_i \sigma_j}.$$

Correlation is useful for interpretation because it normalizes the covariance to be in between -1 and 1. This allows us to compare other investments with each other to see what truly has more "co-movement".

2.2 Vector and Matrix Representation of a Portfolio

Suppose a portfolio contains n assets. Define the vector of portfolio weights

$$\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix},$$

where w_i is the fraction of the portfolio invested in asset i. A fully invested portfolio satisfies

$$\mathbf{1}^T\mathbf{w} = \sum_{i=1}^{n} w_i = 1,$$

where $\mathbf{1}$ is the vector of all ones.

The vector of expected asset returns is

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \end{bmatrix}.$$

The covariance matrix is

$$\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1n} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \cdots & \sigma_{nn} \end{bmatrix},$$

where

$$\sigma_{ij} = \mathrm{Cov}(R_i, R_j).$$

The diagonal entries satisfy

$$\sigma_{ii} = \mathrm{Var}(R_i).$$

This matrix stores the individual variances and all pairwise covariances in one object. It is therefore a way to connect the probability information in the problem to the matrix operations studied in the linear algebra portion of the course.

The covariance matrix also has two important mathematical properties. First, it is symmetric because

$$\mathrm{Cov}(R_i, R_j) = \mathrm{Cov}(R_j, R_i).$$

Second, it is positive semidefinite. For any vector $\mathbf{x}$,

$$\mathbf{x}^T\boldsymbol{\Sigma}\mathbf{x} = \mathrm{Var}(\mathbf{x}^T\mathbf{R}) \geq 0.$$

This explains why the quadratic form used below can represent portfolio variance and can never produce a negative variance.

2.3 Portfolio Expected Return

Let

$$\mathbf{R} = \begin{bmatrix} R_1 \\ R_2 \\ \vdots \\ R_n \end{bmatrix}$$

be the random vector of asset returns. The portfolio return is the weighted sum

$$R_p = w_1 R_1 + w_2 R_2 + \cdots + w_n R_n = \mathbf{w}^T \mathbf{R}.$$

Using linearity of expectation,

$$\begin{aligned} E[R_p] &= E[\mathbf{w}^T \mathbf{R}] \\ &= w_1 E[R_1] + \cdots + w_n E[R_n] \\ &= \mathbf{w}^T \boldsymbol{\mu}. \end{aligned}$$

Therefore,

$$\boxed{\mu_p = \mathbf{w}^T \boldsymbol{\mu}}.$$

This formula is linear in the portfolio weights.

2.4 Portfolio Variance and Diversification

The risk calculation is where covariance becomes especially important. For two random variables, the variance is

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + 2\,\mathrm{Cov}(X, Y) + \mathrm{Var}(Y).$$

Applying the same idea to n assets gives

$$\mathrm{Var}(R_p) = \sum_{i=1}^{n} w_i^2 \sigma_i^2 + 2 \sum_{i<j} w_i w_j \,\mathrm{Cov}(R_i, R_j).$$

In matrix form, the same expression becomes

$$\boxed{\sigma_p^2 = \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w}}.$$

This quadratic form is the central equation in mean-variance optimization [1, 3].

A two-asset case makes the diversification effect easier to see. If a portfolio contains only assets 1 and 2, then

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2 w_1 w_2 \rho_{12} \sigma_1 \sigma_2.$$

If the individual volatilities and weights are held fixed, the final term becomes smaller as correlation becomes smaller. This is why diversification depends on how assets move together rather than simply on the number of assets in a portfolio.

2.5 Global Minimum-Variance Portfolio

The global minimum-variance (GMV) portfolio is the fully invested portfolio with the smallest possible variance. Without restrictions on short selling, the optimization problem is

$$\min_{\mathbf{w}} \quad \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w}$$

subject to

$$\mathbf{1}^T \mathbf{w} = 1.$$

In simpler terms, the vector $\mathbf{w}$ contains the percentage of the portfolio invested in each asset. The expression $\mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w}$ is the portfolio variance derived earlier, so minimizing this expression means finding the combination of asset weights that produces the lowest possible risk. The constraint $\mathbf{1}^T \mathbf{w} = 1$ simply requires all of the portfolio weights to add up to one, meaning that 100% of the available capital is invested. While this notation may look confusing, the idea is simple: we are choosing the portfolio weights w that make the portfolio variance as small as possible, while requiring all of the weights to sum to one.

Using a Lagrange multiplier,

$$\mathcal{L}(\mathbf{w}, \lambda) = \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w} - \lambda(\mathbf{1}^T \mathbf{w} - 1).$$

The Lagrange multiplier allows the variance to be minimized while still enforcing the condition that the portfolio weights add up to one. Here, λ is an additional variable used to incorporate this constraint into the optimization problem. Although Lagrange multipliers were not covered directly in this course, I use the method here as a standard optimization technique to derive the global minimum-variance portfolio.

Taking the derivative with respect to $\mathbf{w}$ gives

$$2\boldsymbol{\Sigma} \mathbf{w} - \lambda \mathbf{1} = 0.$$

Assuming $\boldsymbol{\Sigma}$ is invertible,

$$\mathbf{w} = \frac{\lambda}{2} \boldsymbol{\Sigma}^{-1} \mathbf{1}.$$

Using the constraint $\mathbf{1}^T \mathbf{w} = 1$,

$$1 = \frac{\lambda}{2} \mathbf{1}^T \boldsymbol{\Sigma}^{-1} \mathbf{1}.$$

Therefore,

$$\boxed{\mathbf{w}_{GMV} = \frac{\boldsymbol{\Sigma}^{-1} \mathbf{1}}{\mathbf{1}^T \boldsymbol{\Sigma}^{-1} \mathbf{1}}}.$$

The final equation gives the exact weight vector of the global minimum-variance portfolio. The numerator uses the inverse of the covariance matrix to measure the contribution of each asset to the risk of the portfolio and the denominator normalizes the result such that the sum of all weights equals one. This is the equation I have taken directly into my Python code.

This equation provides a straightforward use of matrix inversion. The course notes show that when an invertible matrix A is present in the equation $A\mathbf{x} = \mathbf{b}$, the answer can be expressed with A^{-1}. The derivation of GMV employs a similar concept as the covariance matrix.

The analytical equation allows for negative weights, which are indicative of short positions. I use this unconstrained solution as a theoretical reference point. I subsequently calculate a long-only GMV portfolio using supplementary limitations.

$$0 \le w_i \le 1.$$

The long-only problem is solved numerically.

The main practical difference between the two versions is that the analytical solution is allowed to assign a negative weight to an ETF while this is not allowed in the long-only version. If you are only long, the weights of the ETFs in the portfolio must be between 0 and 1. That is, you can either hold the ETF or give the ETF a weight of zero, but you cannot short the ETF.

2.6 Efficient Frontier

The GMV portfolio minimizes variance without requiring a particular return. More generally, an investor may want the lowest-variance portfolio that reaches a target expected return μ^*. The optimization becomes

$$\min_{\mathbf{w}} \quad \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w}$$

subject to

$$\mathbf{w}^T \boldsymbol{\mu} = \mu^*,$$

$$\mathbf{1}^T \mathbf{w} = 1,$$

and, in the empirical model,

$$0 \leq w_i \leq 1.$$

Solving the problem for many target returns produces a minimum-variance frontier. The portion at or above the GMV expected return is the efficient frontier because each portfolio on this branch offers the highest expected return available for its level of risk [3, 4].

2.7 Maximum-Sharpe Portfolio

I also compare portfolios using the Sharpe ratio,

$$S(\mathbf{w}) = \frac{\mathbf{w}^T \boldsymbol{\mu} - r_f}{\sqrt{\mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w}}},$$

where r_f is a risk-free rate. For this project, I use a constant annual risk-free rate of 4%. This is a modeling assumption used to illustrate the optimization rather than a quantity estimated from the ETF data.

The maximum-Sharpe portfolio is the long-only portfolio that maximizes $S(\mathbf{w})$. Unlike the GMV portfolio, it considers both expected return and volatility.

3 Data and Implementation

3.1 Dataset

I use daily closing price data from January 1, 2020 through January 1, 2026 for ten commodity ETFs. The data are downloaded from Yahoo Finance which is a Python package `yfinance`.

Table 1: Commodity ETFs used in the analysis

ETF	Main Exposure
GLD	Gold
IAU	Gold
SLV	Silver
PPLT	Platinum
USO	U.S. crude oil
BNO	Brent crude oil
DBA	Broad agriculture
CORN	Corn
CPER	Copper
UNG	Natural gas

For each ETF, I calculate the simple daily return

$$r_{i,t} = \frac{P_{i,t} - P_{i,t-1}}{P_{i,t-1}}.$$

The annualized expected-return estimate is

$$\hat{\mu}_i = 252 \left(\frac{1}{T} \sum_{t=1}^{T} r_{i,t} \right),$$

where 252 is used as the approximate number of trading days in a year. This is as we do not count holidays and weekends as trading days.

This calculation annualizes the arithmetic mean of daily returns. It is an estimate of annual expected return under the assumption that the daily mean is representative, rather than the realized compounded return that an investor would have earned over a particular year.

The annualized sample covariance matrix is estimated as

$$\hat{\mathbf{\Sigma}} = 252\,\widehat{\mathrm{Cov}}(\mathbf{R}).$$

These are sample estimates rather than known population parameters. The optimizer treats them as inputs even though they contain sampling error.

To keep the observations aligned across all ten ETFs, the implementation removes dates with a missing return before estimating the expected-return vector and covariance matrix. This gives every entry of the matrix a common set of dates and avoids comparing covariances estimated from different samples.

3.2 Computational Method

The implementation follows the mathematical framework closely. I first use NumPy to invert the covariance matrix and compute the closed-form unconstrained GMV weights. I then use `scipy.optimize.minimize` which is a python library with the SLSQP method for the long-only problems.

For every long-only portfolio, the constraints are

$$\sum_{i=1}^{10} w_i = 1 \qquad \text{and} \qquad 0 \le w_i \le 1.$$

For the frontier, I generate a range of target returns and minimize portfolio variance for each target. I retain only portfolios at or above the long-only GMV return, which gives the efficient branch. I then solve a separate optimization problem that minimizes the negative Sharpe ratio, which is equivalent to maximizing the Sharpe ratio.

The numerical results can also be checked directly against the constraints. For each reported long-only solution, the weights sum to one, each weight lies between zero and one, and the calculated return and volatility are recomputed from the final weight vector. The optimization status should also indicate successful convergence before a numerical solution is interpreted. These checks are important because a numerical optimizer returns an approximation rather than the closed-form answer available for the unconstrained GMV portfolio.

The covariance matrix deserves an additional numerical check because GLD and IAU track the same underlying commodity and are therefore highly correlated. When two columns of return data are very similar, the covariance matrix can be close to singular, and an inverse can become sensitive to small changes in the data. The long-only optimization reduces the possibility of extreme short positions, but it does not remove estimation uncertainty from the inputs.

The full Python implementation is included in the appendix and is also available in the GitHub repository linked on the title page.

4 Results

4.1 Minimum-Variance Portfolios

Table 2: Summary of optimized portfolios

Portfolio	Expected Return	Volatility	Sharpe Ratio
Unconstrained GMV	11.31%	10.63%	–
Long-only GMV	12.95%	11.49%	–
Maximum Sharpe	16.36%	13.17%	0.9383

The estimated annualized return and volatility of the unconstrained GMV portfolio are 11.31% and 10.63%, respectively. This portfolio allows for negative weights, which represent short positions. This means that the model has more freedom to reduce risk by allowing short selling.

The estimated annualized return and volatility of the long-only GMV portfolio are 12.95% and 11.49%, respectively. Its volatility is a little higher because all weights in the portfolio have to be between 0 and 1. This shows that adding a realistic restriction can change the optimal portfolio even when the expected-return estimates and covariance matrix stay the same.

All three results are in-sample estimates: the same 2020–2025 observations are used to estimate the inputs and evaluate the optimized portfolios. Thus, the reported values reflect the portfolios constructed using this historical sample and should not be taken to represent guaranteed future returns. An out-of-sample test would be required to evaluate how the allocations would perform on data not used by the optimizer.

4.2 Correlation and Diversification

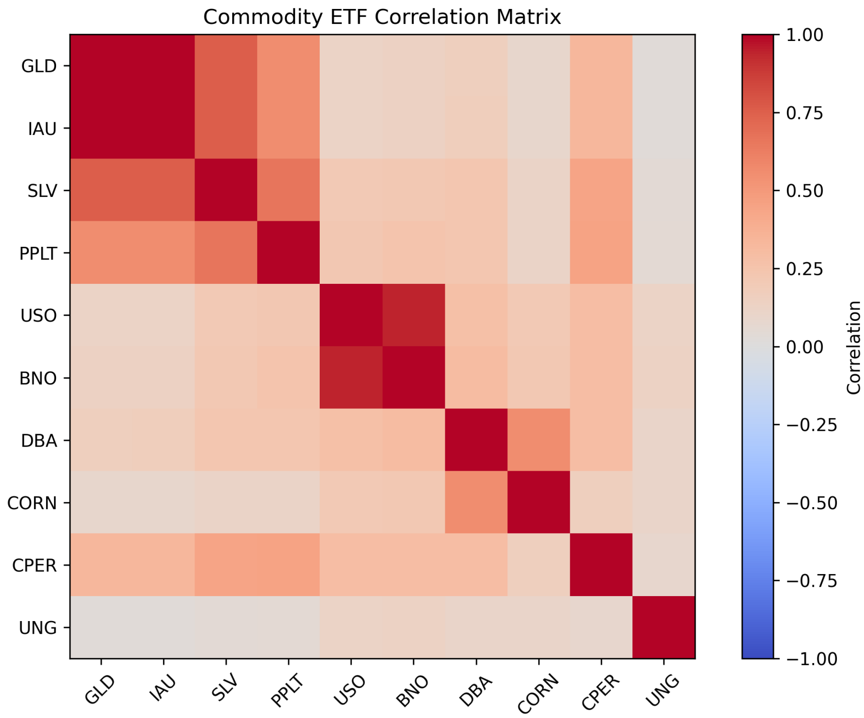

Figure 1: Correlation matrix of daily commodity ETF returns

Figure 1 shows how closely the daily returns of the ETFs moved together during the sample period. GLD and IAU have a strong positive correlation because both funds track gold. USO and BNO are also strongly correlated because both provide exposure to crude oil. In comparison, ETFs from different commodity groups generally have lower correlations.

These relationships help explain the benefits of diversification. Combining assets with lower correlations can reduce portfolio variance because their returns do not always move in the same direction. Therefore, diversification depends not only on the number of assets in a portfolio, but also on the covariance between them.

4.3 Efficient Frontier

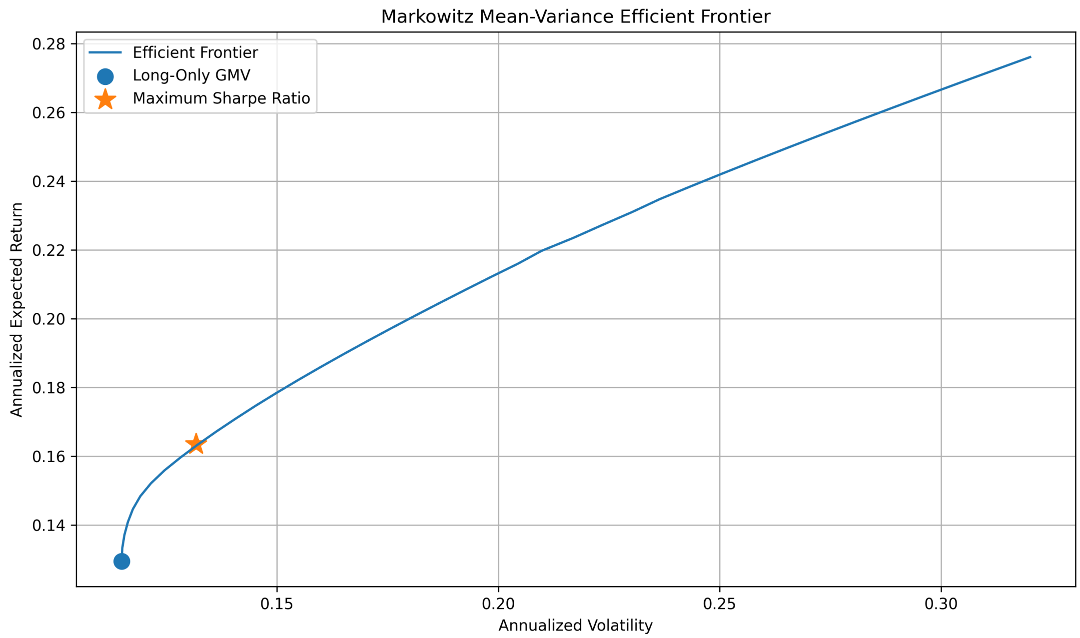

Figure 2: Long-only Markowitz efficient frontier

Figure 2 shows the relationship between annualized expected return and annualized volatility. The long-only GMV portfolio is located at the left end of the frontier because it has the lowest volatility among the long-only portfolios. Moving upward along the frontier gives a higher expected return, but it also requires taking on more risk.

The maximum-Sharpe portfolio lies farther along the frontier because it does not minimize risk alone. Instead, it selects the portfolio with the highest excess return per unit of volatility. Its expected return is 16.36%, its volatility is 13.17%, and its Sharpe ratio is 0.9383.

Portfolios to the right of the efficient frontier are inefficient because another feasible portfolio can provide the same expected return with less volatility. The frontier therefore represents the best risk-return combinations produced by the model.

4.4 Maximum-Sharpe Portfolio

In this case, assuming a risk-free rate of 4%, the maximum-Sharpe portfolio has an estimated annualized return of 16.36%, an annualized volatility of 13.17%, and a Sharpe ratio of 0.9383.

This portfolio is different than the minimum variance portfolio because the objective is different. The GMV portfolio ignores expected return and minimizes variance only. The maximum-Sharpe portfolio takes on more risk when the gain in expected return is sufficient to increase the ratio of excess return to volatility.

A useful lesson of this comparison is that the word "optimal" only has sense after an objective has been specified. The portfolio that minimizes risk optimally may not maximize risk-adjusted returns optimally.

5 Limitations

We find the model useful but with a number of assumptions that constrain the literalness of the results.

The first step is to estimate expected returns and covariances from historical data. These estimates are then used as inputs to the optimization, assuming that future commodity returns may not follow the same pattern as the 2020-2025 sample. Sample mean returns can be unstable and optimized weights can be sensitive to relatively small changes in the estimates.

The second is that the model's definition of risk is variance. Variance is symmetric in the sense that it treats positive and negative deviations from the mean in the same way but investors are usually more concerned about large unexpected losses than large unexpected gains.

Third, the empirical model does not consider transaction costs, taxes, bid-ask spreads, management fees, and turnover constraints. This long-only constraint is still a simplification but it makes the model more realistic than the unconstrained solution.

Fourth, the maximum-Sharpe calculation assumes that the risk-free rate is 4% per year. If you change the assumption, the Sharpe ratios will change and the maximizing portfolio may change.

Lastly, these ETFs are not perfect substitutes for the underlying physical commodities. Returns of some funds that gain exposure thru futures contracts can also be affected by the futures markets and the rolling process.

These limitations do not make mean variance optimization obsolete. Instead, they argue that the model should be viewed as a mathematical framework for structuring the trade-off between risk and return, rather than as a sure-fire way of predicting the optimal future portfolio.

6 Conclusion

This project applied ideas from the class to a practical portfolio problem. The probability side begins by treating asset returns as random variables. Expected value describes average return, variance measures dispersion, and covariance measures how pairs of returns move together. These quantities then become the inputs to the portfolio model.

Linear algebra makes it possible to organize the same information compactly. Portfolio weights form a vector, expected returns form another vector, and all pairwise covariances form a covariance matrix. With this notation, expected portfolio return becomes

$$\mathbf{w}^T\boldsymbol{\mu},$$

while portfolio variance becomes

$$\mathbf{w}^T\boldsymbol{\Sigma}\mathbf{w}.$$

The unconstrained global minimum-variance portfolio also gives a direct application of matrix inversion through

$$\mathbf{w}_{GMV} = \frac{\boldsymbol{\Sigma}^{-1}\mathbf{1}}{\mathbf{1}^T\boldsymbol{\Sigma}^{-1}\mathbf{1}}.$$

The empirical analysis uses historical commodity ETF returns to estimate the expected-return vector and covariance matrix. The long-only efficient frontier shows the tradeoff between expected return and volatility, while the correlation matrix makes the role of diversification visible. Comparing the GMV and maximum-Sharpe portfolios also shows that different definitions of "best" can produce different allocations.

The main takeaway is that diversification is fundamentally a covariance problem. Simply adding more assets does not automatically create a well-diversified portfolio. What matters is how those assets move relative to one another. For me, this was the clearest connection between the two parts of the course: probability describes the uncertainty in returns, while linear algebra provides the structure for combining those relationships into one portfolio model.

A Code

The code and research for this project are available in the following GitHub repository. I recommend visiting the repository for a clearer picture of the project's implementation and methodology:

github.com/kshoop28/mean-variance-portfolio-optimization

The complete Python implementation is available at the repository linked above.

B Bonus

I have also created many other projects in quantitative finance in the past. They can be found on my GitHub portfolio at github.com/kshoop28.

References

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